Maulik C. Madhavi

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Experience

NCS PTE LTD, Singapore, Snr Product Engineer (Video Analytics Researcher) — Apr 2021 – present, 3 years 4 months

- Innovate the video analytics algorithm of person/pedestrian attribute, face attributes
- Developed the video analytics algorithm for recognition, detection and inference for generic computer vision training. To support backend engine for inhouse KAI CC product.
- Explore the generative AI capabilities to automate video analytics using cutting edge large language model and vision langauge model technologies.

National University of Sinagpore (NUS), Singapore, Research Fellow — Dec 2017 – Apr 2021, 3 years 4 months

- Support research and project activities. Involved in autonomous bus chatbot project, healthcare ASR system and SCDF automatic form filling system using intent and entity extraction.
- Mentor and co-supervise the final year undergrad students for their projects.

Education

Ph.D. — **Dhirubhai Ambani Institute of Information and Communication Technology (DA-IICT), Gandhinagar.**, Information and Communication Technology (ICT) — July 2011 – Nov 2017

- Thesis title - 'Design of QbE-STD System - audio representation and matching perspective'

Master of Technology (M.Tech) — **Dhirubhai Ambani Institute of Information and Communication Technology (DA-IICT), Gandhinagar.**, Information and Communication Technology (ICT) — July 2009 – June 2011

- Communication Systems specialization
- CPI - 8.26/10
- Thesis title - 'Person recognition from their hum'

Bachelor of Engineering (BE) — **L.V.P. Engineering College, Saurashtra University, Rajkot**, Electronics and Communication (EC) — July 2005 – June 2009

- Percentage (69.73

Publications

Vocal Tract Length Normalization using a Gaussian mixture model framework for query-by-example spoken term detection — November 2019

M. C. Madhavi, H. A. Patil

10.1016/j.csl.2019.03.005 (Computer, Speech and Language, Elsevier)

Design of mixture of GMMs for query-by-example spoken term detection — November 2018

M. C. Madhavi, H. A. Patil

10.1016/j.csl.2018.04.006 (Computer, Speech and Language, Elsevier)

Combining evidences from magnitude and phase information using VTEO for person recognition using humming — November 2018

H. A. Patil, ***M. C. Madhavi***

10.1016/j.csl.2018.04.006 (special issue of recent advances in speaker and language recognition and characterization Computer Speech and Language, Elsevier)

Partial matching and search space reduction for QbE-STD September 2017

M. C. Madhavi, H. A. Patil

10.1016/j.csl.2018.04.006 (Computer, Speech and Language, Elsevier)

Static and dynamic information derived from source and system features for person recognition from humming

H. A. Patil, ***M. C. Madhavi***, K. K. Parhi

Int. Journal of Speech Technology, Springer

Additional Experience And Awards

Instructor (2003-2005): Taught 2 full-credit computer science courses

Third Prize, Senior Design Project: Awarded 3rd prize for a synchronized calendar project out of 100 entries

Technologies

Languages: C++, Python

Tools: Bash, Git, Docker, Latex

Research Interests: Audio processing, Computer Vision, Application of generative AI in computer vision